Digital Holography : numerical simulation and reconstruction

Lab 1 Discrete Fourier Transform and spectrum analysis

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Python will be used for the simulation, display and processing of physical optical signals. The Python functions to be used are given in the lab text. You will have to read the help for their syntax (use the contextual help window).

For each lab session, a fill-in-the-blank code will be provided to you to start the practical work. You have to rename it (SURNAME_Lab1.ipynb) and publish a report that will be uploaded on Moodle.

It is mandatory to annotate/comment your code.
Two lab reports **must** be uploaded on Moodle : one at the **end of the session** and another one (completed) **before the next session**. It will contain the programs/functions and the images results as well as your comments. It should be personal.

« +++ » questions are optional.

0. Preliminaries

The material of this lab session is on Moodle in the folders : **"Lab0 to begin" and "Lab1_DH" .**
Download this material.
and follow the instruction given by the professor.
(1) Open the folder "**Lab0 to begin**"

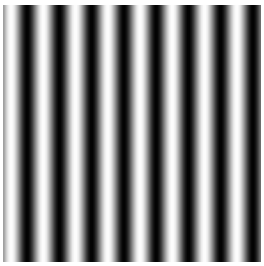
- run the code “Lab0_Basics.ipynb”
  to learn/remind the basics of Python
- run the code “Lab0_Main_example.ipynb”
  to learn how to load/display images, program functions in Python, print a report, ...

(2) Read the text of the lab ("**Lab1_DH_text.pdf**").
(3) In the folder **Lab1_DH,** a code to begin the lab is given: **"Lab1_DFT.ipynb".**
The images are in the subfolder **"Lab1_images"**

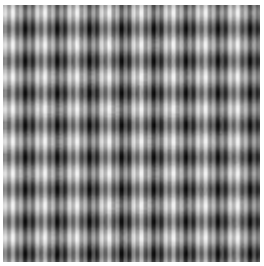
At the end of the lab, publish your report in *pdf* and upload it on Moodle.

The report must contain:
the code
the plot and display of images
your answers to questions and your comments.

1. 2D Discrete Fourier Transform (DFT)



sinus.png



sinus2.png

1 Display of the image "sinus.png" and its spectrum

a. Load in the workspace the image sinus.png ('cv.imread(path,cv.IMREAD_GRAYSCALE)' see Lab0_Main_example). Convert in double type the image matrix (`np.double`). Display the image (`plt.imshow`).
Add labels to the abscissa and ordinate axis (the units must always be given, here they are "x(pix)", "y(pix)").

b. Compute its spectrum (*fft2*).

c. Compute and display its phase and its modulus (*angle, abs*)
By default the origin of the frequencies axis is on the left top corner and we visualize frequencies: $0 \leq \mu_x < 1$, $0 \leq \mu_y < 1$). We have to center the origin using *fftshift* in order to visualize $-1/2 \leq \mu_x < 1/2$, $-1/2 \leq \mu_y < 1/2$).
Understand and comment the code.

d. Let's consider that pixel width is 5 µm. Display again the spectrum using frequency units in m^{-1}.

2 Spectrum analysis

a. Analyze the spectrum modulus (*abs*) and its phase (*angle*) (symmetries, location of the peaks, peaks amplitude, ...).

b. +++ The Fourier transform is invertible: test it by computing an inverse FT of the spectrum (*ifft2*).

c. What happen in space domain if the magnitude of the zero frequency is put to zero in Fourier domain ?
Try it using the following code.

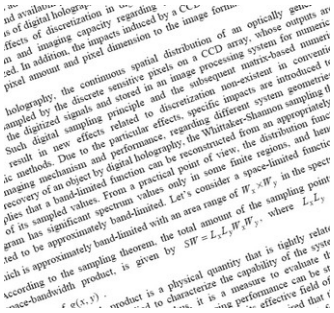
```
FFT_I_0=np.fft.fft2(I)
FFT_I_0[0,0]=0
I_0=np.real(np.fft.ifft2(FFT_I));
```

d. What happen if all the values of the spectrum phase are put to zero before inverse FT ?

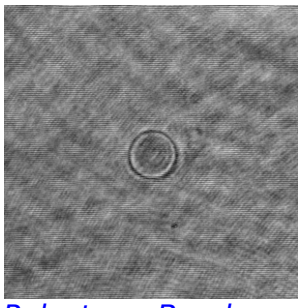
3 +++ (optional) Do the same work on "*sinus2.png*".

4 +++ Display the image "*Spectre3.png*". Create an image which modulus would approximately be equal to "*Spectre3.png*". Are there several solutions ? Why ?

2. Spectrum analysis



Text.png



PolystyrenBead.png



ImSattNewYork.png

1 Measurements of periods

a. Display the image "text.png" and its spectrum.
b. What are the differences between the two displays (classical one, and logarithm one) ?
c. Analyze the spectrum and measure the lines periodicity. Check that your measurement is correct in the image.
d. If the peak in the Fourier domain is located in the spectrum with an accuracy of ± 0.5 pixels in both x and y directions, what is the accuracy of the measured period?
e. Measure the rotation angle of the text.
f. +++What other measurements can be performed ?

2 The image "*PolystyrenBead.png*" is an interferometric image. The pixel size is 2.2 µm.
a. Using the image (space domain) measure the period of the carrier fringes.
b. Make the same measurement using the spectrum (Fourier domain).
c. Compute the accuracy of each measurement in space and frequency domain. Discuss on the results.

3 ++ Same questions on the image «*ImSattNewYork.png* ».

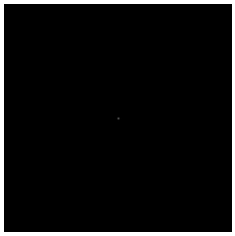
4 +++ Select an image of your choice. Display its spectrum. Measure its periodicities.

Notes : To better visualize a spectrum modulus
- suppress its (0,0) frequency
or
- display its logarithm (do not forget to add a constant value to avoid computing log(0)),
or
- use a color LUT (plt.colorbar())
or
- use a surface display of a 2D signal:
see Lab0_Main_example

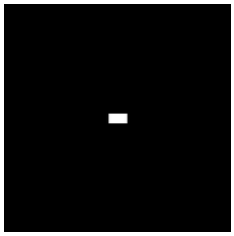
3. +++ Fourier Transforms of commonly used functions

Visualize the modulus of the spectrum of commonly used sampled functions (for a better visualization, we will use a color LUT (*colormap, colorbar*) and/or a surface display (e.g. see illustration of the lecture):

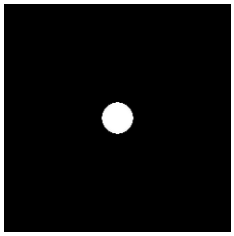
- Dirac (1 single pixel is lightened)
- 2D rectangular function
- Disk function
- 2D comb function
- White noise



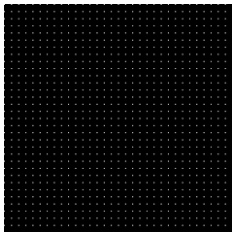
pixel.png



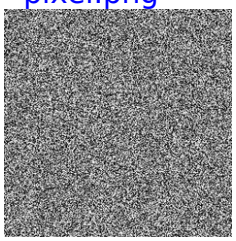
Rect2D.png



Disk.png



Comb2D.png



WGN.png